

Nguyen Ngoc Minh Khanh

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SUMMARY

I am an AI Engineer and MLOps Specialist with over five years of experience supporting AI-driven initiatives within software development environments. My work focuses on AI research, experimentation, and integrating modern AI tools to enhance developer productivity and deliver meaningful business outcomes. I'm currently completing my Master's in Artificial Intelligence at VNUHCM – University of Science.

I enjoy collaborating in cross-functional teams to explore, pilot, and implement practical AI solutions. Committed to continuous improvement, I prioritize defining clear success metrics, regularly assessing impact, and promoting a culture of experimentation to help engineering teams adopt effective AI strategies.

SKILLS RELEVANT

- **Languages:** Python, Java*, Bash*
- **Frameworks:** PyTorch, TensorFlow, Kubeflow, Spark, Polar
- **Cloud/DevOps:** Message queues, Docker, Jenkins, K8S, GCP
- **AI Engineer/MLOps Engineer:** Expertise in designing, optimizing, deploying diverse Machine Learning/Deep Learning models and orchestrating end-to-end MLOps pipelines

EXPERIENCE

- **Knorex - Advertising domain** Fulltime
Senior AI Engineer *Dec 2024* *Feb 2023 - Present*
 - Optimized the CTR data pipeline using Spark on Dataproc, cutting execution time by 3x.
 - Enhanced CVR pipeline performance by migrating from Pandas to Lazy Polars, achieving a 30x reduction in execution time.
 - Drove significant cost savings by reducing BigQuery usage and restructuring MongoDB, shrinking database size from 3TB to 500GB, resulting in \$300 monthly savings.
 - Mentored an intern on the company's tech stack and workflows, guiding them through the development of data pipelines and machine learning models for click fraud detection.
MLOps Engineer II *Feb 2023 - Dec 2024*
 - Collaborated closely with AI and NLP teams to develop advanced metrics, delivering critical insights into business dynamics and model performance.
 - Led the end-to-end lifecycle of CTR and CVR models, driving architecture, optimization, and the creation of high-performance training pipelines. Ensured smooth deployment with TensorFlow Serving and flawless integration with Bidopt.
 - Developed a click-quality decision tree model to detect and flag click spikes, improving system accuracy.
 - Teamed up with system engineers to optimize model deployment and developed a proactive monitoring system using Prometheus and Grafana dashboards, enabling real-time tracking and continuous performance evaluation.
Data Engineer *Feb 2023 - Aug 2023*
 - Spearhead the maintenance of data engineering pipelines, ensuring constant accuracy and availability of data.
 - Innovate and construct new pipelines for cutting-edge features like Shopping Max and Performance Max, prioritizing high delivery standards.
- **Vinbigdata** Fulltime
AI Engineer - build virtual assistant for electric cars *Dec 2020 - Feb 2023*
 - **Text Normalization & Inverse Text Normalization System:**
 - * Researched, designed, and constructed a Text Normalization system for a Vietnamese and English Text To Speech product: speed and accuracy increased by > 50% and > 40%, respectively.
 - * Maintained, and developed an Inverse Text Normalization system for the Automatic Speech Recognition product.
 - * Built and deployed internal service by docker.
 - **Automatic Speech Recognition:**
 - * Utilized the Kaldi framework to create Automatic Speech Recognition for production in both Vietnamese-English.
 - * Built a substantial general and domain language model by creating and normalizing more than 100GB of text data.

- * The model is built by > 5 terabytes of processed audio data by multi-threaded programming, guaranteed the latency, and served the model for streaming prediction using Websocket API.

- **Deep Noise Suppression:**

- * Research techniques to help remove noise, enhance audio quality, and improve speech recognition in backend.
- * Built the entire SOTA Speech Enhancement model from beginning to end, guarantee of latency for downstream tasks, such as call bots and smart speakers.

- **FPT Telecom**

Fulltime

Data Science Intern

Jul 2020 - Dec 2020

- **Event classification:** Built fault event classification models with an accuracy of over 90% after performing data analysis

- **Singularity**

Parttime

Software Engineer Intern - eKyc system for banking

Mar 2020 - Jul 2020

- **Detect true corner cut for a digital forensic process:** Image processing with OpenCV v.3.4 and Deep Learning Yolo v.3: achieved more than 80% accuracy in the laboratory.
- **L&D Postgresql Administration:** server administration such as backup and restore, monitoring, high availability architecture, master-slave replication.

- **Freelancer**

Parttime

Software Engineer Role

Nov 2019 - Apr 2020

- **Room rental app:** Database design, MVC-based restful API implementation, deployed on GCP using Node.js backend
- **Cinema backend Nodejs:** Front-end pages were produced using HTML, CSS, and react.js. Used a NoSQL database with a powerful querying and Restful API on the Google Cloud Platform.

PROJECTS

- **Talent Acquisition Search:** Built an intelligent search and response application for Talent Acquisition, leveraging AWS cloud services, and harnessing the power of generative AI for conversational interaction. Key technologies: LLM, RAG, AWS bedrock, Elastic Search.
- **Code Understanding:** A web application that allows users to upload a code repository and chat with an AI assistant to understand and explore the codebase. Key technologies: LLM, Langchain, RAG.
- **Thesis - Integrating WordNet into neural machine translation for English to Vietnamese(scored 10/10):** Research-oriented, built a 2-way machine translation system in English and Vietnamese using Deep Learning technique. Tech: Python, Pytorch, Fairseq Toolkit.

HONORS AND AWARDS

- **Honors:** VinIF Master's Scholarship — Vingroup Innovation Foundation (VinIF), Vietnam *2023–2025.*
- **Paper:** Cross-Architecture Knowledge Distillation for Speech Enhancement: From CMGAN to UNet, *published to Q1 Journal Neurocomputing, 2025.*
- **Paper:** FastSpeechStyle: Fast, Emotion Controllable, and High-Quality Speech Synthesis. *International Journal of Asian Language Processing*, vol. 32, no. 02n03, pp. 2350007, 2022. World Scientific.
- Huawei Seeds for the Future Program, 2020.
- 1st Place in Data science track of Developer Circle Vietnam Innovation, 2019.

EDUCATION

- **VNUHCM - University of Science** Ho Chi Minh, Vietnam
Master of Artificial Intelligence, 2022 – Aug 2025 — Current GPA: Distinction
- **VNUHCM - University of Science** Ho Chi Minh, Vietnam
Bachelor of Technology - Computer Science; GPA: 3.25

REFERENCES

Nguyen Tien Huy - Lecturer & Lead Research Scientist Vinbigdata
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